Devinder **Kumar**

ML Systems Engineering Leader

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Summary

ML engineering leader specializing in the architecture, lifecycle management, and operationalization of global AI/LLM platforms within complex, large-scale production environments. Proven track record in scaling multi-disciplinary engineering teams and building secure, self-service infrastructure necessary to deploy foundational models with strict cost and data governance guardrails. Expert in MLOps, LLMOps, and driving enterprise-wide operational excellence at scale

Work Experience

TD Bank (AI Platform & Layer6)

Toronto, ON

Head of ML Systems & Engineering, AI Platform — *Nov 2024 - Present*

Head of ML Product Engineering, Layer6 — *Apr 2022 - Oct 2024*

- Head of ML Systems & Engineering group at TD Bank, leading a **32-member engineering org** split into four dedicated teams (GenAI foundation, ML Standardization, LLMOps/MLOps, and Platform Engineering), each managed by its own engineering leader.
- Spearheaded the design & governance of a centralized, self-service LLM delivery framework, empowering cross-functional business lines to seamlessly discover, provision, & consume open-source & commercial APIs at scale. **Decreased PTU usage by 50%** & improved latency by 66%
- Established enterprise-wide LLMOps best practices encompassing automated failover, secure data egress controls, and cost-performance optimization (including endpoint usage/PTU monitoring) for multi-region model deployment.
- Architected and implemented a **global enterprise MLOps target operating model**, standardizing secure, multi-region AI delivery and platform governance across diverse lines of business in Canada, the US, and Singapore.
- Leading engineering strategy and implementation of ML engineering framework & processes for TD's enterprise wide AI Platform to realize the enterprise's ambition of deploying 100s of models on public cloud using Databricks, Kubernetes etc., for ML model deployment at scale.
- Led creation of standardized ML workflows & operations resulting in a **75% improvement in time to market** from 6 months to 1.5 months.
- Led the successful implementation of the first customer-facing ML model in TD bank, first ML predictive tabular foundation model-TD AI Prism & spearhead the development of new enterprise frameworks to operationalize & serve ML models at scale.
- Responsible for operations for a portfolio of ML models (100+) in production that are delivering **$145M annually**.
- Grew the ML systems & engineering org from 7 to 32 by hiring 15 new ML engineers and consolidating leadership of MLOps, Infra & DevOps team.

Senior Engineering Manager — *Jun 2021 - Apr 2022*

- Within a year, **doubled the number of ML models in production** on the Azure cloud infrastructure.
- Led the team through leadership change & navigated capacity issues while expanding the ML engineering team from 4 to 7 members.
- Senior EM for ML Engineering team responsible for data exploration, co-developing ML models with ML Scientists, model production & maintenance.

Next AI

Toronto, ON

Lead AI Scientist — *Jan. 2017 - Jan. 2020*

- Scientific mentor for 7 AI/ML start-ups in the NextAI accelerator's inaugural cohort (20 startups) including the winner of the best startup award.
- Within a year, promoted to lead AI scientist managing a team of 4 scientist and continued to mentor a total of 18 AI startups in my tenure.

Machine Learning Consultant

Toronto, ON

Lead Consultant — *Sept 2015 - Feb 2020*

- Led, advised, and implemented 5 AI based PoC solutions for various large and mid-sized corporations in manufacturing, mining, automation, via UWaterloo & independent projects. All the clients were F500, multi $B+ annual revenue companies.
- Examples include: PoC for an automation company for the detection of people in a protected area and integration with the airport PLC system & PoC for one of the largest mineral mining companies to find excavation sites.

Philips Research (HQ)

Eindhoven, Netherlands

Deep Learning Intern — *Aug 2016 - Dec 2016*

- Developed a state-of-the art visualization technique to visualize the decision making of deep learning networks and find biases in the learned models. Later, presented the paper at 2017 NeurIPS workshop for Machine Learning for Health (MLH).
- Developed the first internal visualization (explainable) library for deep learning neural networks to make it available for use for Data Scientists.
- Gave an invited talk at Bayer's international conference of computational science in Berlin (Germany) on visualizing neural network decisions.

Education

Stanford University

Stanford, CA, USA

Postdoctoral Fellow at Stanford Cognitive & Systems Neuroscience Laboratory (SCSNL) — *Mar 2020 - Jun 2021*

- Advisor: Prof. Vinod Menon at School of Medicine, Stanford University.
- Co-authored 2 journal papers and supervised one Stanford undergraduate research assistant over 8 months

University of Waterloo & Vector Institute *Waterloo & Toronto, Canada*

PhD in Systems Design Engineering *Jan 2017- Dec 2019*

- Dissertation Topic (Explainable AI): Class Based Strategies for Understanding Neural Networks
- Advisor: Dr. Alexander Wong (Canada Research Chair, Medical Imaging & AI) and Dr. Graham Taylor (Canada Research Chair-ML , CIFAR AI Chair, Research Director Vector Inst.)
- Multiple year Ontario Graduate Scholarship (1/1 in Engineering Department & 1/5 in UWaterloo) & President's Graduate Scholarship

Honors & Awards

2026	**American Banker - Innovation of the Year (Honor List),** TD AI Prism	*New York, USA*
2025	**Big Innovation Awards - AI business impact,** AI Platform	*Toronto, CA*
2023	**AI Impact Helix Award-TD,** Real-time pre-mortgage approval	*Toronto, CA*
2017	**Best Paper Award,** Interpretable ML Workshop at 31st Neural Info Processing & Systems (NeurIPS)	*Long Beach, CA*
2013	**Best Paper Award,** IEEE Region 10 Paper Contest across Asia-Pacific Region	*Seoul, S.Korea*
2012	**World Semi-finalist,** Dell Social Innovation Challenge	*Austin, USA*
2018,'19	**Award,** President Graduate Scholarship (PGS) & International Ontario Graduate Scholarship (OGS)	*Waterloo, ON*
2016	**Award,** Magna Cum Laude, Annual Meeting of the Imaging Network of Ontario	*London, ON*

Media Coverage

TD Announces Launch of Groundbreaking Predictive Foundation Model

Yahoo Finance *June 2025*

Better way found to determine the integrity of metals

EurekAlert (AAAS), Phy.org *July 2018*

This Researcher Wants to Open the 'Black Box' of Financial AI

VICE's Motherboard *Oct 2017*

Scientists developed software to make artificial intelligence systems more trustworthy

Technology.org *Oct 2017*

Building trust in AI

Investment Executive *Sept 2017*

Presentation

Big Data Workshop *Belfast, N.Ireland*

Invited talk at the **Ulster University** *Apr 2021*

- What clinicians should know about deep learning and explainable AI

Element AI (Acquired by Service-Now) *Toronto & Montreal, Canada*

Invited talk *Feb 2019*

- Explainable AI - Techniques, Challenges and beyond Interpretability

Fritz-Haber-Institut der Max-Planck-Gesellschaft *Berlin, Germany*

Invited Talk *Dec 2016*

- Visualizing the Decision Making Process of CNNs

Gustave Roussy Cancer Research Institute *Paris, France*

Invited Talk *May 2016*

- Discovery Radiomics for Cancer detection: Intelligent bio-markers

Publication

Publication Summary

Patents (filed): 45, Papers: 34 (16 Journal articles, 18 conference papers)
Citations: 2050, H-index: 16, I-index: 18
Full list here: `https://tinyurl.com/5aab5rws`